

China Pharma and Biotech

Asia Biopharma Summer Teach-in: GLP-1 and new approaches to metabolic management



Rebecca Liang, Ph.D.
+852 2123 2656
rebecca.liang@bernsteinsg.com



Ellie Li
+852 2123 2621
ellie.li@bernsteinsg.com



Miki Sogi, Ph.D.
+81 3 6777 6991
miki.sogi@bernsteinsg.com



Tian Tan
+81 3 6777 6976
tian.tan@bernsteinsg.com

The obesity market is entering its next phase, where competition extends beyond weight loss efficacy. While next-generation injectable therapies have pushed placebo-adjusted weight loss beyond 20%, differentiation is increasingly shifting toward broader metabolic benefits, lean mass preservation, and convenience. We believe future winners will not only deliver superior weight reduction but also address cardiovascular, renal, liver, and body composition outcomes.

Oral GLP-1s and long-acting therapies are expanding the treatment toolbox. While injectable therapies continue to provide the strongest efficacy, oral GLP-1s are creating a compelling alternative for patients seeking easier administration and moderate weight loss. Lilly's Foundayo (orforglipron) and Novo Nordisk's oral semaglutide have demonstrated that oral therapies can achieve approximately 10% weight loss, which may be sufficient for many patients. Foundayo's small-molecule design provides additional convenience because it avoids the strict fasting requirements associated with peptide-based oral semaglutide. Among China-based programs, Innovent's IBI3032 has emerged as a notable contender after delivering placebo-adjusted weight loss of 8.6% in only four weeks, while weekly oral candidate IBI3042 highlights the industry's broader push toward reducing dosing frequency without sacrificing efficacy.

Emerging modalities aim to extend durability beyond traditional GLP-1 therapies. Antibody-peptide conjugates (APCs), circRNA therapeutics, and gene therapies are being developed to achieve monthly dosing or potentially long-lasting metabolic control following a single administration. Amgen's MariTide provides the most advanced proof-of-concept today, while Chinese innovators including Innovent and PegBio are actively building exposure to these next-generation platforms.

Despite pricing pressure, we continue to project a ~CNY85bn China GLP-1 market by 2030E. Despite accelerated price erosion and increasing regulatory scrutiny around obesity-drug prescribing, underlying demand remains robust. We have revised our market assumptions to reflect lower long-term pricing but higher penetration and a broader eligible population, resulting in an unchanged estimate of approximately CNY85bn for the China GLP-1 market by 2030E. A major launch wave beginning in 2027 should further expand patient access and accelerate market penetration. In our view, the industry's next stage of growth will be driven by a combination of efficacy, metabolic benefits, body composition, convenience, and affordability rather than any single attribute alone.

Note: Eli Lilly and Amgen are covered by Courtney Breen, Novo Nordisk is covered by Justin Smith.

BERNSTEIN TICKER TABLE

Ticker	Rating	23 Jul 2026		Price Target	TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
		Cur	Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
9926.HK (Akeso)	M	HKD	102.80	130.00	(56.6)%	CNY	(0.60)	(0.52)	0.70	(146.9)	(172.1)	127.1
ONC (BeOne)	O	USD	319.93	412.00	(10.3)%	USD	2.63	5.65	8.93	121.6	56.6	35.8
1093.HK (CSPC)	M	HKD	8.37	10.70	(30.5)%	HKD	0.37	0.52	0.56	19.6	14.0	12.9
3692.HK (Hansoh)	O	HKD	32.58	44.00	(34.0)%	CNY	0.93	0.97	0.88	30.3	29.1	32.0
1801.HK (Innovent)	O	HKD	91.05	120.00	(24.8)%	HKD	0.48	0.71	2.90	187.7	128.3	31.4
600276.CH (Hengrui)	O	CNY	54.91	65.00	(32.2)%	CNY	1.18	1.42	1.78	46.5	38.8	30.8
6990.HK (Kelun-Biotech)	O	HKD	521.50	526.00	10.8%	CNY	(1.66)	(2.75)	0.85	44.9	45.6	28.1
1177.HK (Sino BioPh)	M	HKD	5.19	7.90	(50.7)%	CNY	0.19	0.20	0.22	23.4	22.0	20.2
9688.HK (Zai Lab)	M	HKD	16.03	15.00	(73.8)%	USD	(0.16)	(0.15)	(0.09)	0.5	0.5	0.4
ASIA			1,907.19									
SPX			7,498.96									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

6990.HK, 9688.HK valuation is EV/Sales (x); 9926.HK, 1093.HK, 1177.HK base year is 2024;

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

We rate BeOne, Hansoh, Innovent, Hengrui, and Kelun-Biotech Outperform, and Akeso, CSPC, SBP, and Zai Lab Market-Perform.

DETAILS

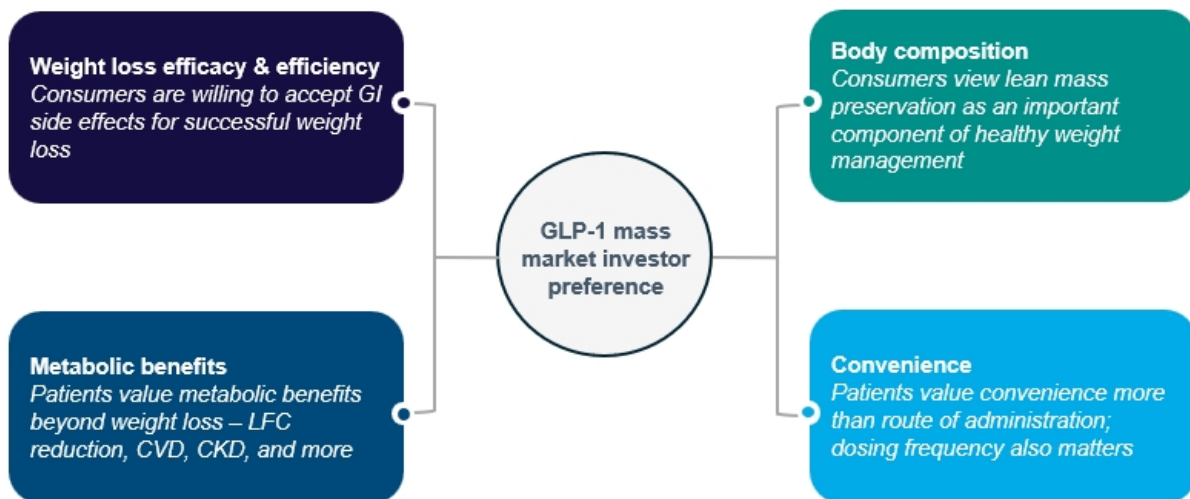
Traditionally, discussion around GLP-1 therapies has been dominated by clinical efficacy, safety, and commercial adoption. However, as the category expands beyond early adopters and moves toward a broader obesity and metabolic health market, it is equally important to understand how consumers themselves evaluate these products. To prepare for this GLP-1 teach-in, we reviewed a range of patient and consumer preference research and identified several themes that consistently emerge as the most important factors influencing treatment choice.

First, weight-loss efficacy remains the single most important attribute. Multiple studies suggest that patients are willing to tolerate, at least in the short term, gastrointestinal side effects if doing so translates into meaningfully greater weight loss. In other words, once safety is considered acceptable, efficacy often becomes the primary determinant of preference. This reflects the reality that many patients view weight reduction as the core therapeutic objective and are willing to make trade-offs to achieve better outcomes ([source](#)).

Second, consumers are increasingly focusing on metabolic and body composition benefits. Broader metabolic and body composition benefits are receiving increased attention. Metrics such as lean mass preservation, visceral fat reduction, and overall quality of weight loss are becoming important differentiators as consumers seek healthier and more sustainable outcomes rather than simply maximizing pounds lost ([source1](#), [source2](#)).

Finally, convenience. Oral therapy has been an evolution after traditional subcutaneous forms, and companies are also exploring long-acting agents. Based on these themes, our GLP-1 teach-in will focus on these key areas that may shape the next phase of competition within the GLP-1 landscape.

EXHIBIT 1: **Weight loss, convenience, body composition/metabolic benefits are the key investor preference when choosing GLP-1 products**

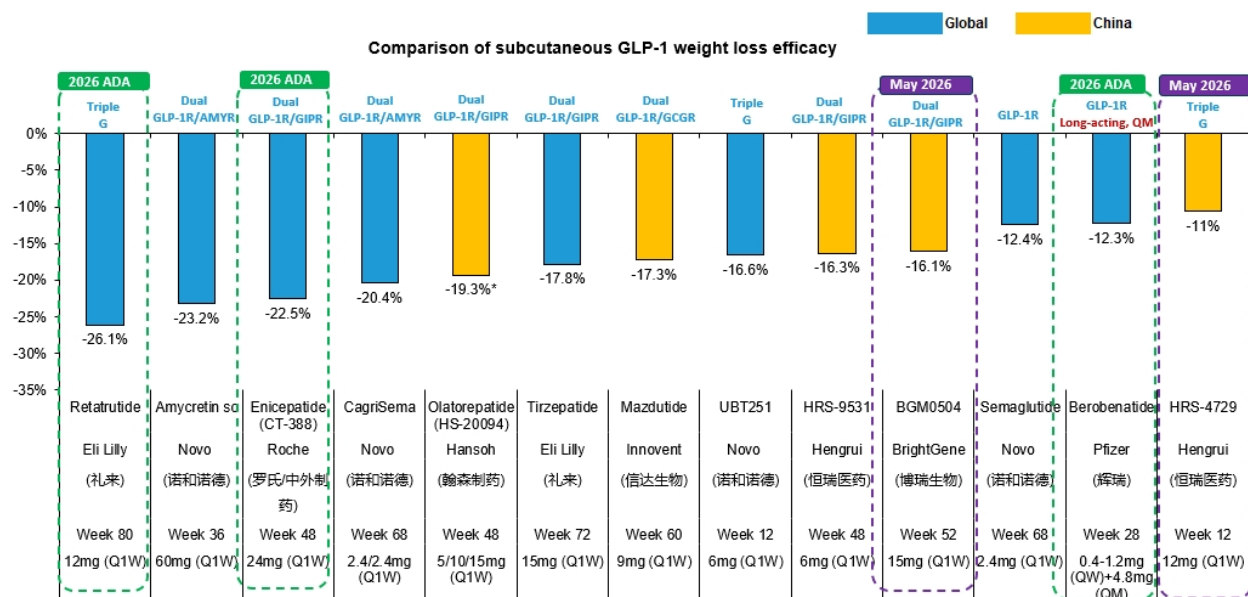


Source: Diabetes Obes Metab, Patient Preference and Adherence, Scandinavian Journal of Primary Health Care, Patient Care, Bernstein analysis

PART 1: WEIGHT LOSS EFFICACY AND EFFICIENCY

Efficacy benchmark in weight loss has evolved from -10% to more than -20% after placebo adjustment. Only a few years ago, the market viewed achieving more than 10% placebo-adjusted weight loss as a significant breakthrough, with semaglutide setting a new standard at approximately 12.4%. Since then, the pace of innovation has accelerated dramatically. Tirzepatide pushed placebo-adjusted weight loss close to 18%, while a new wave of next-generation candidates - including retatrutide, subcutaneous amycretin, enicepatide (CT-388), and CagriSema—have demonstrated placebo-adjusted weight loss exceeding 20% in clinical studies. As a result, the competitive landscape is entering a new phase where multiple therapies are capable of delivering similar levels of substantial weight reduction.

Note: Novo Nordisk and Roche are covered by Justin Smith; Eli Lilly and Pfizer are covered by Courtney Breen

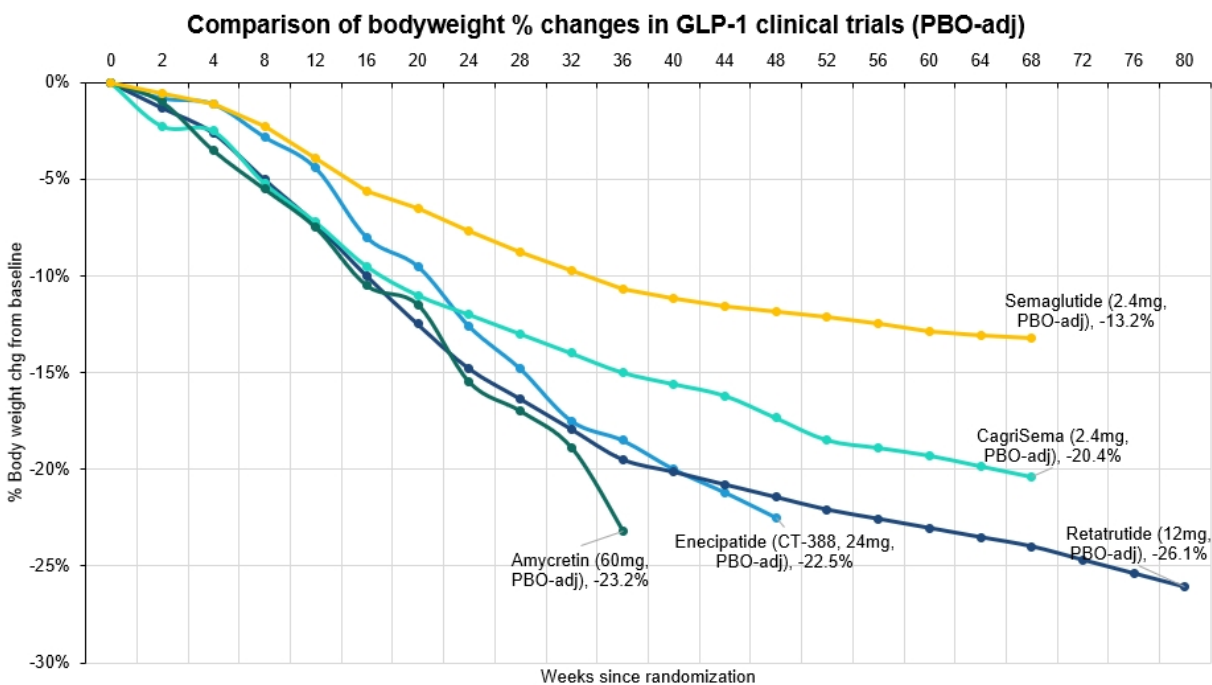
EXHIBIT 2: **Competitive landscape of subcutaneous GLP-1**

1) Mazdutide's GLORY-2 (9mg) and CagriSema's REDEFINE-1 results at 2026 ADA are the same as previous reportings; 2) HS-20094: Likely not PBO-adj; Eli Lilly and Pfizer are covered by Breen Courtney, Novo Nordisk and Roche are covered by Justin Smith, BrightGene is not covered
Source: Company disclosure, ADA, EASD, Bernstein analysis

Among the key contenders, retatrutide delivered approximately 26.1% placebo-adjusted weight loss by Week 80, while subcutaneous amycrin and enicepatide (CT-388) achieved around 23.2% and 22.5%, respectively, by Weeks 36–48. CagriSema also demonstrated strong efficacy at roughly 20.4% placebo-adjusted weight loss by Week 68. Importantly, the differentiation is not limited to final weight-loss outcomes. **The weight-loss trajectories show that these agents begin separating from one another relatively early in treatment**, with some candidates demonstrating double-digit weight loss by approximately 16–24 weeks and continuing to widen the gap thereafter. This suggests that the new generation of therapies is not only achieving greater absolute weight loss, but also delivering meaningful efficacy at a faster pace (Exhibit 3).

Compared with semaglutide, which achieved approximately 12.4% placebo-adjusted weight loss at Week 68, the leading next-generation agents begin to show visible differentiation as early as the first few weeks of treatment and increasingly so by the 10–20 week timeframe. Mechanistically, this is consistent with the industry's shift from single-target GLP-1 agonism toward multi-target approaches that engage complementary metabolic pathways, such as GLP-1/GIP, GLP-1/amylin, or even triple G mechanisms. These therapies therefore appear to represent more than an incremental improvement on semaglutide; they reflect a broader enhancement in the intensity and breadth of metabolic regulation. From a commercial perspective, the earlier onset of efficacy may also be an important advantage. Patients who experience noticeable weight loss within the first several weeks of treatment are more likely to perceive the therapy as working, which could support better adherence, persistence, and ultimately long-term treatment retention.

EXHIBIT 3: Leading subcutaneous GLP-1s in the space achieve >20% weight loss, demonstrating superior efficacy from the early stages of treatment compared to semaglutide



Source: Company disclosure, NEJM, Lancet, Bernstein analysis

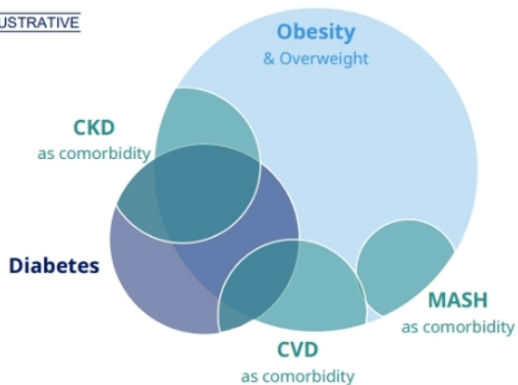
PART 2: METABOLIC BENEFITS

There is substantial patient overlap across obesity, diabetes, CKD, CVD, and MASH / MAFLD, with these conditions often coexisting as manifestations of shared underlying metabolic dysfunction. Obesity is a known risk factor and a common comorbidity for these diseases. For example, about [44%](#) of individuals with CKD also have obesity, and [39%](#) of obesity patients have MAFLD. Therefore, treatment of obesity with GLP-1 drugs have resulted in metabolic benefits that alleviated other diseases as mentioned (Exhibit 4).

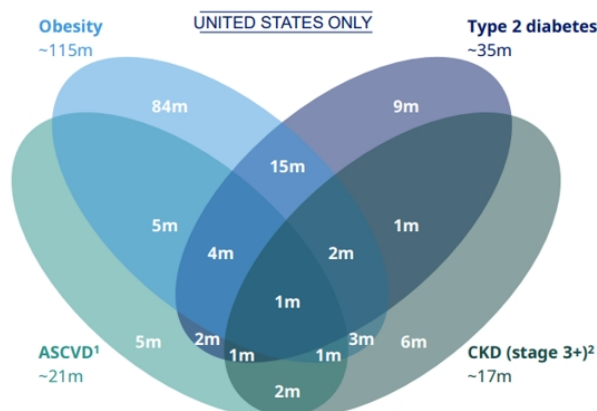
EXHIBIT 4: Large overlapping population between obesity, diabetes, CKD, CVD, and MASH

Focus will remain on core therapy areas and prioritising unmet needs, including comorbidities

ILLUSTRATIVE



Patient overlaps for key focus areas in type 2 diabetes



Source: Novo Nordisk

The obesity market is increasingly evolving beyond weight loss alone, with GLP-1 therapies demonstrating benefits across a growing range of comorbidities including cardiovascular disease (CVD), chronic kidney disease (CKD), heart failure, MASH, and obstructive sleep apnea (OSA). **Semaglutide has established the broadest evidence base to date, with approved indications and positive outcomes across cardiovascular, renal disease, and MASH**, while tirzepatide has generated particularly strong data in MASH and OSA. These findings reinforce the view that obesity is a systemic disease and that GLP-1 therapies can deliver clinically meaningful benefits across multiple organ systems, potentially expanding the addressable patient population well beyond those seeking weight reduction alone.

Looking ahead, next-generation agents are aiming to further differentiate through enhanced efficacy and broader cardiometabolic benefits. Among the emerging therapies, mazdutide and retatrutide stand out, particularly in liver-related outcomes, with placebo-adjusted reductions in liver fat content of approximately 81% and 83%, respectively—substantially greater than those reported for semaglutide or tirzepatide. Retatrutide has also shown promising data in OSA and osteoarthritis, while several ongoing Phase 3 outcomes studies will determine whether these efficacy advantages translate into superior cardiovascular and renal protection. As the field matures, competition is likely to shift from weight loss alone toward a broader assessment of total metabolic benefit across obesity-related complications.

EXHIBIT 5: **Approved indications: CVD, CDK, MASH, OSA, and more players coming into the space**

	Trial completed and indication approved	Trial with positive data but not yet approved	Ongoing trials pending result						
Medication	Other metabolic benefits								
	CVD	HFpEF	CKD	UA reduction	MAFLD / MASLD	MASH	OSA	OA	AD
Semaglutide	(-20%/26% MACE risk)	(-7.8 in KCCQ-CSS)	(-24% kidney disease)		(-42% LFC reduction)	Fibrosis improvement: 37% vs 22% (PBO) pts MASH resolution: 63% vs 34% (PBO) pts	(-52% in OSA incident)	(-14.2 WOMAC)	-
	SELECT/SUSTAIN 6 (Ph3)	STEP HFpEF (Ph3)	FLOW (Ph3)		NCT02970942 (Ph2)	ESSENCE (Ph3)		STEP 9 (Ph3)	
Tirzepatide	-	(-6.9 in KCCQ-CSS)	-	about -54µmol/L, ~48 after PBO-adj	(-47% LFC reduction)	Fibrosis improvement: 55% vs 30% (PBO) pts MASH resolution: 62% vs 10% (PBO) pts	(-48% AHI)	-	-
	SURMOUNT-MMO (Ph3)	SUMMIT (Ph3)	TREASURE-CKD (Ph2)		SYNERGY-NASH (Ph2)	SYNERGY-NASH (Ph2)	SURMOUNT-OSA (Ph3)		
Mazdutide	(CVD benefits)	-	(kidney benefits)	about -37µmol/L, ~45 after PBO-adj	(-81% LFC reduction)	-	-	-	-
		NCT06862908 (Ph2)			GLORY-3 (Ph3)	NCT06937749 (Ph2)	GLORY-OSA (Ph3)		
Retatrutide	-		-		(-83% LFC reduction)	-	(-45% AHI)	(-35% WOMAC)	-
	TRIUMPH-Outcomes (Ph3)		TRIUMPH-Outcomes (Ph3)		SYNERGY-Outcomes (Ph3)	SYNERGY-Outcomes (Ph3)	TRIUMPH-1 (Ph3)	TRIUMPH-1/4 (Ph3)	
Oral semaglutide	(-14% MACE risk)	-	-	about -12µmol/L	-	-	-	-	Unsuccessful
	SOUL (Ph3)								EVOKE (Ph3)

Results are PBO-adj. except for MASH patients response; OA=osteoarthritis, OSA=obstructive sleep apnea, AD=Alzheimer's disease, CVD=Cardiovascular disease, HFpEF=heart failure with preserved ejection fraction, CKD=Chronic kidney disease

Measurement metrics: KCCQ-CSS=Kansas City Cardiomyopathy Questionnaire-Clinical Summary Score; AHI=Apnea-Hypopnea Index; WOMAC=Western Ontario and McMaster Universities Osteoarthritis Index

Source: Company disclosure, NEJM, Lancet, Clinicaltrials.gov, Bernstein analysis

Beyond weight loss, GLP-1-based therapies may provide a range of important metabolic and cardiovascular benefits.

In the pooled mazdutide group, participants experienced significantly greater improvements across several cardiometabolic risk factors compared with placebo after 48 weeks of treatment. Mazdutide reduced systolic blood pressure by 8.56 mmHg, compared with a 2.13 mmHg reduction in the placebo group, resulting in a treatment difference of -6.44 mmHg. Similar benefits were observed for lipid parameters, including triglycerides, total cholesterol, and LDL cholesterol, all of which declined to a greater extent with mazdutide than with placebo.

In addition, mazdutide demonstrated favorable effects on metabolic and liver-related biomarkers. Serum uric acid levels decreased by 36.7 µmol/L in the mazdutide group versus an increase of 7.91 µmol/L with placebo, while alanine aminotransferase (ALT), a marker of liver function, showed a substantially greater reduction with mazdutide. These findings suggest that the clinical benefits of GLP-1 receptor agonist therapies extend beyond weight reduction alone, contributing to broader improvements in cardiovascular, metabolic, and hepatic health (Exhibit 6).

EXHIBIT 6: **Other metabolic benefits**

Table 3. Key Secondary End Points in the Pooled Mazdutide Group, as Compared with Placebo (Treatment-Policy Estimand Analysis).*			
Variable	Mazdutide, Pooled (N = 405)	Placebo (N = 205)	Difference
No. of participants with data at wk 48	374	196	—
Change from baseline to wk 48 (95% CI)			
Systolic blood pressure — mm Hg	−8.56 (−9.90 to −7.23)	−2.13 (−3.78 to −0.48)	−6.44 (−8.22 to −4.65) †
Triglycerides — mmol/liter	−0.65 (−0.77 to −0.54)	−0.16 (−0.28 to −0.03)	−0.50 (−0.64 to −0.36) †
Total cholesterol — mmol/liter	−0.28 (−0.38 to −0.18)	0.15 (0.04 to 0.27)	−0.43 (−0.56 to −0.31) †
LDL cholesterol — mmol/liter	−0.20 (−0.28 to −0.12)	0.10 (0.02 to 0.19)	−0.31 (−0.41 to −0.20) †
Serum uric acid — μmol/liter	−36.70 (−46.01 to −27.39)	7.91 (−2.76 to 18.58)	−44.61 (−56.56 to −32.65) †
Alanine aminotransferase — U/liter	−13.92 (−16.45 to −11.39)	−4.44 (−7.43 to −1.44)	−9.48 (−12.75 to −6.21) †

Source: NEJM, Bernstein analysis

PART 3: BODY COMPOSITION

Lean mass preservation as the core differentiation beyond GLP-1s. Looking ahead, the next phase of innovation in the GLP-1 landscape is increasing defined by lean mass preservation rather than weight loss alone. Amylin therapies have emerged as a key differentiator approach, delivering reasonable weight loss while demonstrating better preservation of lean mass, with a fat-to-lean mass ratio of c.3:1 — materially better than conventional GLP-1s. Mechanistically, amylin’s gut-brain signaling allows patients to retain food enjoyment before satiety, potentially avoiding the nausea and food aversion commonly seen among GLP-1 therapies. Beyond amylin, targets within the activin pathway are gaining traction: ActRII class (such as bimagrumab & LAE102) has shown the ability to induce reasonable weight loss alongside absolute gains in lean mass; on the other hand, INHBE/ALK7-related therapies (such as ARO-INHBE & WVE-007) appear even more skewed toward lean mass accretion. However, though bimagrumab achieves weight loss with lean mass gains, it is an IV treatment, and Eli Lilly terminated a phase 2b obesity trial before enrolling patients for strategic business reasons in 2025, though bimagrumab is still “currently being studied in an ongoing phase 2 trial with tirzepatide in patients with obesity” ([source](#)).

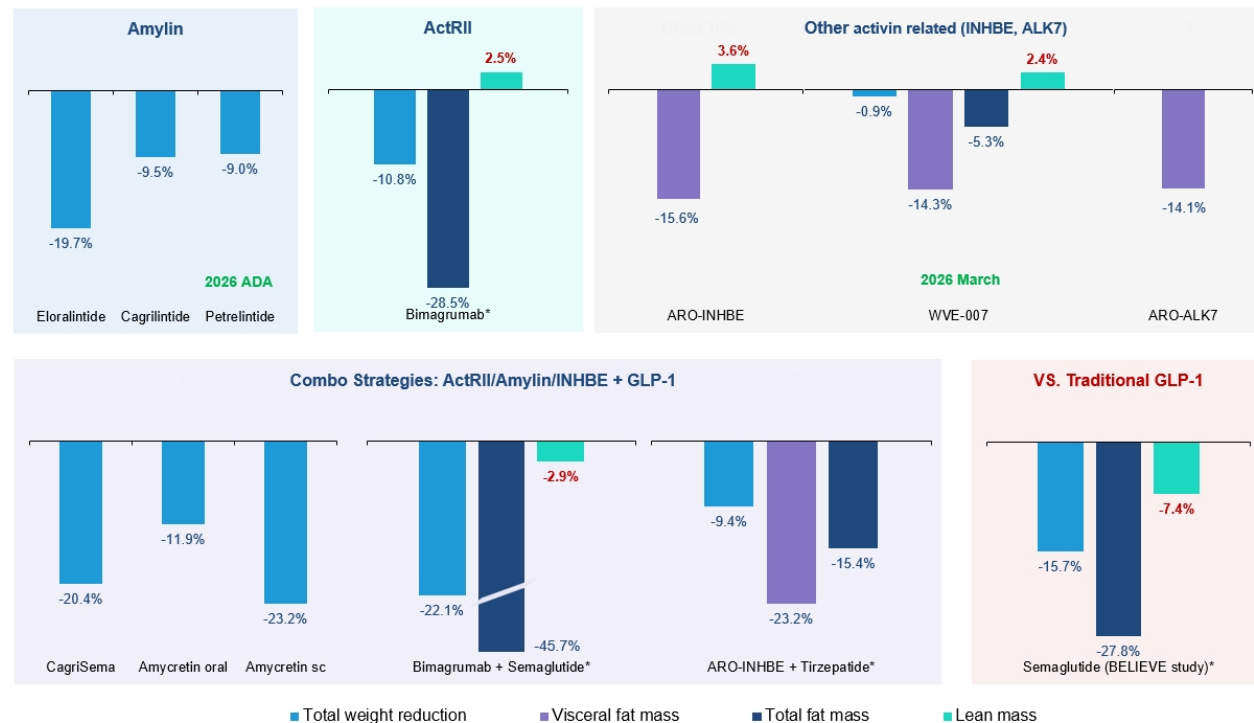
Expanding the efficacy frontier: combo, oral, and triple-agonist strategies. In parallel, multiple strategic directions are being explored to extend the efficacy and usability of GLP-1s. Combination regimens pairing GLP-1s with complementary mechanisms (including amylin, ActRII, and INHBE) aim to achieve more pronounced weight loss (potentially exceeding 20%) while partially mitigating lean mass erosion. Oral formulations, including Wegovy pill and orfo, are increasingly positioned for long-term maintenance, addressing adherence and convenience in chronic weight management. Meanwhile, triple-G (GLP-1/GCCR/GIP) agonists offer the promise of the most substantial weight loss; however, this benefit is accompanied by a higher expectation of lean mass loss.

EXHIBIT 7: **Emerging obesity therapies offer a wide spectrum of personalized weight loss options**

Approach	Asset	Dosing	Administ ration	Max weight loss	Weight loss range	Change in lean mass	Lifestyle intervention ?
Oral	Wegovy pill, Orfoglipron	QD	PO	12%	8 - 12%	Loss of lean mass	Yes (Both)
Triple G	Retatrutide, UBT251	Q1W	SC	27%	17 - 27%		Yes (Retatrutide)
Amylin	Eloralintide, Cagrilintide	✓ Q1W	SC	20%	10 - 20%	Around 3:1 fat-to-lean mass reduction ratio	Yes (Both)
Activin (ActRII)	Bimagrumab	Q4W/(Q12W in this combo trial)	IV	11%	11%	Gain lean mass by 3%	-
INHBE/ALK7 (also activin related)	WVE-007, ARO-INHBE/ALK7	≥ Q4W (WVE-007 single dose observed at 3m)	SC	1%	1%	Gain lean mass by 1-4%	No (WVE-007)
Combo strategy	Amylin + GLP-1 (CagriSema, Amycretin)	-	SC / PO	23%	12 - 23%	Loss of lean mass potentially mitigated (Bima + Sema combo was -3%)	Yes (CagriSema)
	ActRII + GLP-1 (Bimagrumab + semaglutide)	-	SC + IV	22%	22%		-
	INHBE + GLP-1 (ARO-INHBE + Tirzepatide)	-	SC	9%	9%		-

✓ means moderate weight loss and balanced preservation of lean mass

Source: Company disclosure, Lancet, ADA, Bernstein analysis

EXHIBIT 8: **Emerging therapies work better when combining with GLP-1 treatments**

Names with * are not PBO-adj

Source: Company disclosure, Lancet, ADA, Bernstein analysis

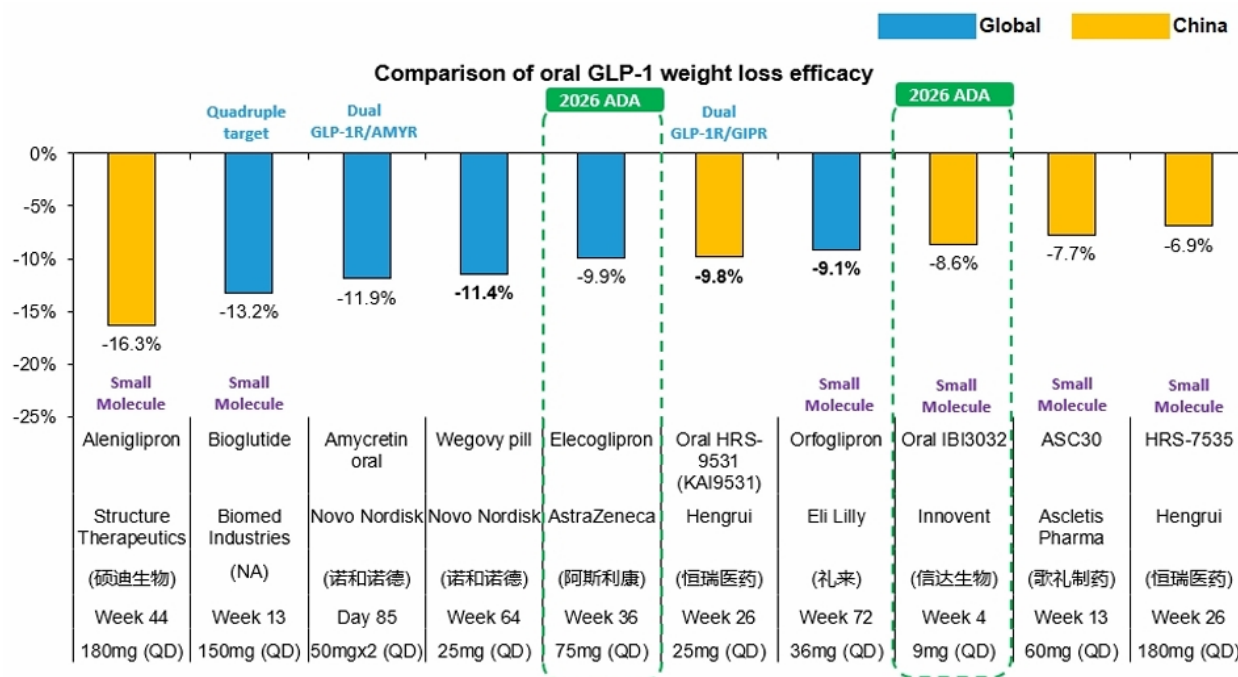
PART 4: CONVENIENCE

Injectable GLP-1 therapies remain the most effective option for weight loss, with leading products delivering mid- to high-teens percentage reductions in body weight. **Oral therapies generally offer somewhat lower efficacy, but recent data suggest many candidates can still achieve around 10% weight loss.** As a result, oral agents may be a good option for patients who are not targeting the highest levels of weight reduction but value the convenience of a daily pill over injections.

Among oral therapies, Foundayo (orforglipron) is differentiated by its small-molecule design. While oral semaglutide (Wegovy pill) is a peptide that must be taken under strict fasting conditions, Foundayo does not require the same dietary restrictions and can be taken more conveniently. This simpler administration, combined with weight-loss efficacy in the ~10% range, could make Foundayo an attractive alternative for patients seeking ease of use rather than maximum weight-loss efficacy.

Another oral GLP-1 candidate attracting attention is Innovent's IBI3032. Although still in early-stage development, the daily oral small-molecule therapy delivered an impressive placebo-adjusted 8.6% weight loss in just four weeks, a magnitude of effect that rivals what some competitors achieve only after many months of treatment. While cross-trial comparisons should be interpreted cautiously and longer-term data are still needed to assess durability, the rapid onset of weight loss seen with IBI3032 suggests it could emerge as a highly competitive entrant in the oral obesity market if subsequent studies confirm these early results.

EXHIBIT 9: **Competitive landscape of oral GLP-1 weight loss efficacy**

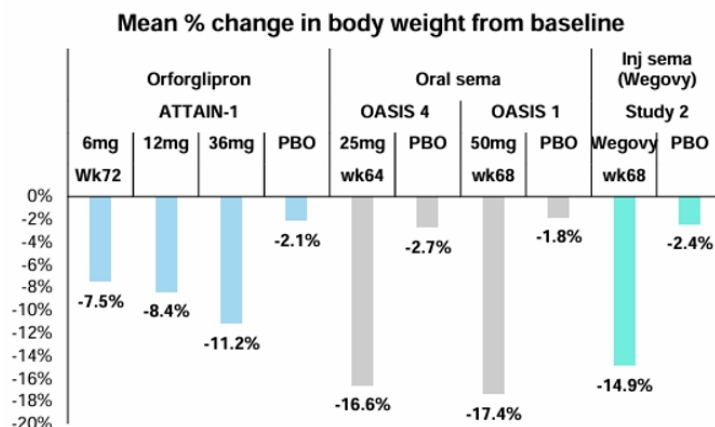


Results of oral HRS-9531, ASC30, orfoglupron, and aleniglipron at 2026 ADA are the same as previously reported; AstraZeneca is covered by Justin Smith, Structure Therapeutics, Biomed industries, and Ascleptis Pharma are not covered

Source: Company disclosure, ADA, NEJM, Bernstein analysis

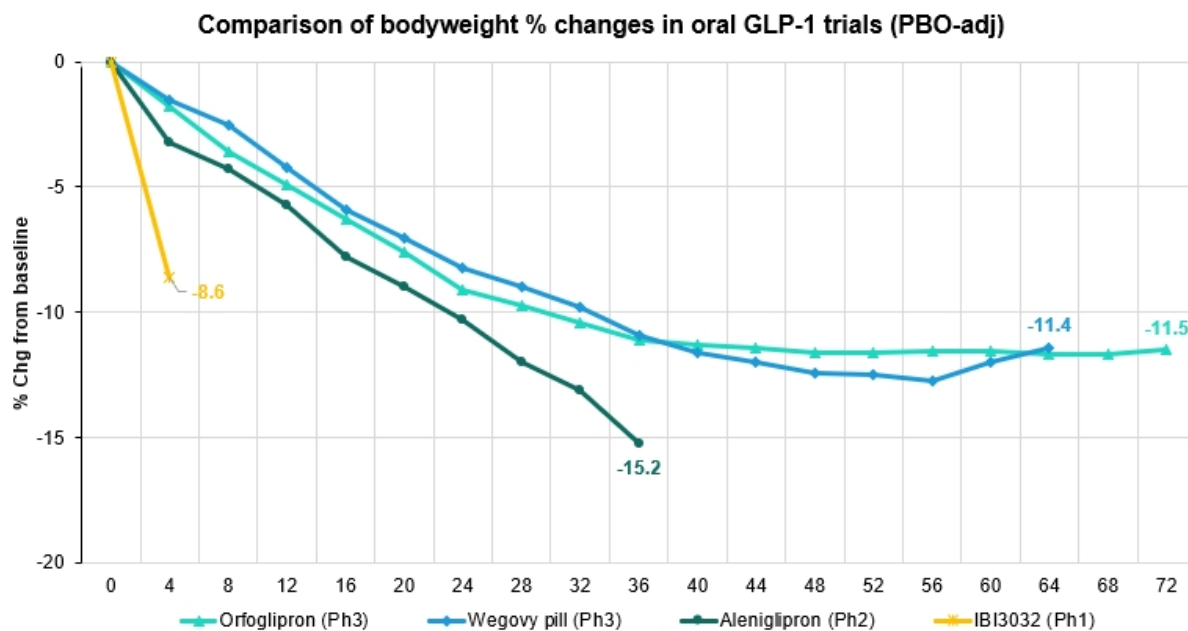
EXHIBIT 10: Oral GLP-1's battle has started between Foundayo and Wegovy pill

Product	Foundayo	Wegovy Pill
Generic name	orforglipron	semaglutide
Company		
Modality	Small molecule	peptide
Dietary restriction	None	Take the med with empty stomach and no drink or eat for 30 mn



Source: Company reports, The Lancet, NEJM, Bernstein analysis

EXHIBIT 11: IBI3032, an oral GLP-1 drug of Innovent, achieved PBO-adj weight loss of -8.6% in 4 weeks



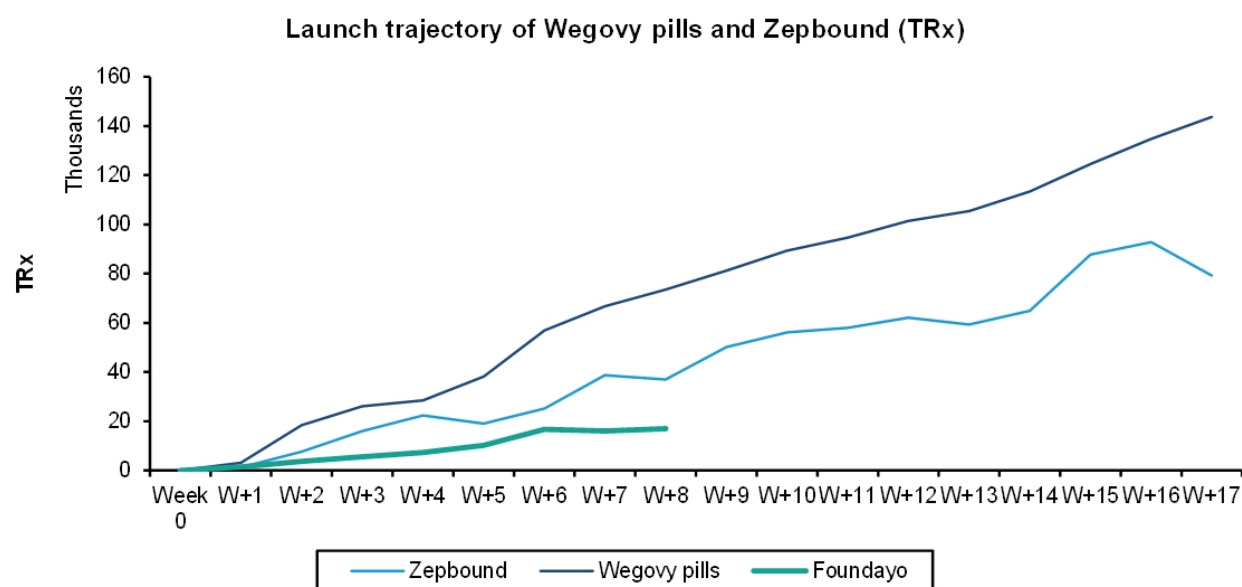
Source: NEJM, ADA, Bernstein analysis

EXHIBIT 12: **Oral GLP-1 weight loss efficacy by subgroup**

Drug / Formulation	Dose & regimen	Time point	Weight loss %	≥10% weight loss (% of participants)
Oral HRS-9531 (GLP-1/GIP)	10 mg QD	Week 26	-6.7%	59.1%
	25 mg QD		-11.9%	
	50 mg QD		-11.4%	
	Placebo		-2.1%	
IBI3032 oral (GLP-1 small molecule)	8mg QD	Week 4	-8.8%	
	9mg QD		-10.7%	
	Placebo		-2.1%	
Orforglipron oral (GLP-1 small molecule)	6 mg QD	Week 72	-7.5%	33.3%
	12 mg QD		-8.4%	40.0%
	36 mg QD		-11.2%	54.6%
	Placebo		-2.1%	12.9%
Oral semaglutide (GLP-1 small molecule)	25 mg QD	Week 64	-13.6%	63.0%
	Placebo		-2.2%	14.4%

Source: Company disclosure, ADA, NEJM, Bernstein analysis

Despite its launch earlier this year and differentiated convenience profile, **Foundayo's prescription uptake has so far trailed that of Wegovy pills**. As our analyst Miki Sogi noted, one potential reason is familiarity: oral semaglutide has been available for years, giving physicians and patients greater comfort with its efficacy and safety profile, whereas Foundayo is a new entrant that is still building awareness. This suggests that Lilly may need to invest further in patient and physician education to help translate its convenience advantages into stronger prescription growth.

EXHIBIT 13: **Foundayo's launch has been much slower than Wegovy Pill's: one of key differences is the latter was a known entity from Day 1**

Source: Company reports, IQVIA, Bernstein analysis

Looking beyond daily oral therapies, Innovent is also developing IBI3042, a novel once-weekly oral small-molecule GLP-1 receptor agonist. Preclinical data suggests that IBI3042 can achieve weight-loss effects comparable to Lilly's daily

oral therapy orforglipron, while offering the added convenience of once-weekly dosing. Although these findings are still at the animal-study stage and will need to be validated in human trials, IBI3042 highlights the industry's efforts to combine the convenience of oral administration with less frequent dosing, potentially narrowing the gap between oral and injectable obesity treatments.

EXHIBIT 14: IBI3042 at 2026 ADA

P: OBESITY—ANIMAL | JUNE 05 2026

2543-P: IBI3042: A Novel Oral Once-Weekly Small-Molecule GLP-1 Receptor Agonist for Type 2 Diabetes and Obesity **FREE**

YE JIANG; JIALI DING; YAO XIONG; SHILAN LIU; DECHENG REN

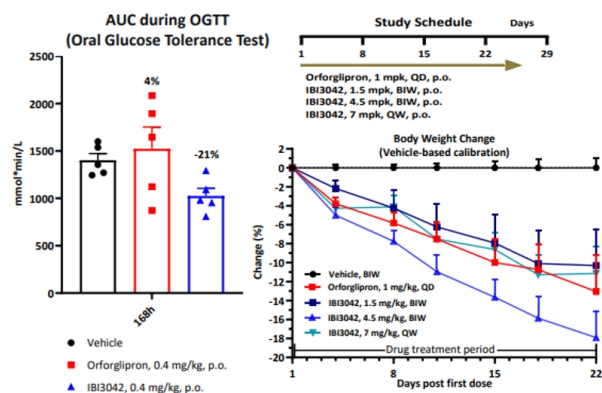


Diabetes 2026;75(Supplement_1):2543-P

<https://doi.org/10.2337/db26-2543-P>

Source: ADA, Bernstein analysis

EXHIBIT 15: IBI3042 achieves comparable efficacy compared with orforglipron



Source: Innovent

Long-acting GLP-1 therapies are emerging as an important next-generation trend in obesity treatment, with developers aiming to extend dosing intervals from weekly injections to once-monthly (Q4W) or even quarterly administration. The rationale is straightforward: reducing dosing frequency could improve patient adherence and convenience while maintaining meaningful weight-loss efficacy. Among the global programs, Amgen's MariTide and Pfizer's beronematide (MET-097i) are the most advanced, both currently in Phase 3 development. The highlighted products have also generated the most compelling efficacy data so far, suggesting that less frequent dosing does not necessarily come at the expense of efficacy. **China-based companies are also actively pursuing the long-acting GLP-1 opportunity.** QL Biopharm's ZT002 stands out among domestic programs, reporting approximately 12% placebo-adjusted weight loss at week 24 with once-monthly dosing, placing it broadly in line with leading global assets. In contrast, earlier-stage programs such as Raynovent's RAY1225 and Ascletis' ASC30 have shown more modest efficacy to date, with placebo-adjusted weight loss of 9.4% and 6.3%, respectively. While cross-trial comparisons should be interpreted cautiously, current data suggest that beronematide and ZT002 represent the leading efficacy profiles within the long-acting GLP-1 landscape and may set the benchmark for future ultra-long-duration obesity therapies.

Note: Amgen and Pfizer are covered by Courtney Breen

EXHIBIT 16: **Long-acting GLP-1s: Monthly GLP-1s are approaching the efficacy of leading weekly therapies**

Company	Product	Target	Modality	Dosing frequency	Progress	
					Phase (Primary completion)	Data (PBO-adj)
Global						
Amgen	MariTide	GLP-1R agonist /GIPR antagonist	Peptide-Antibody Conjugate	Q4W	Phase 3 (2027-01)	WL at week 52: -13.7%
Pfizer	Berobenatide (MET-097i)	GLP-1R agonist	Peptide	Q4W	Phase 3 (2028-05)	WL at week 28: -12.3%
	MET-815i (MET-097i's prodrug)	GLP-1R agonist	Peptide	Quarterly	Phase 1 (-)	-
Domestic						
CSPC	SYH2082	GLP-1R/GIPR agonist	Peptide	Q4W	Phase 1 (2026-10)	-
	SYH9017	Semaglutide biosimilar	Peptide	Q4W	Phase 1 (2026-08)	-
QL Biopharm	ZT002	GLP-1R agonist	Peptide	Q4W	Phase 3 (2027-05)	WL at week 24: -12%
Qilu/Raynovent	RAY1225	GLP-1R/GIPR agonist	Peptide	Q2W	Phase 2 (2025-12)	WL at week 24: -9.4%
Ascletis	ASC30	GLP-1R agonist	Small molecule	Q4W	Phase 2 (2026-03)	WL at week 12: -6.3%

Highlighted products show better efficacy

Source: Company disclosure, ADA, Bernstein analysis

Beyond longer-acting peptides and small molecules, the next wave of obesity innovation is exploring entirely new modalities designed to extend durability, reduce dosing frequency, and potentially improve efficacy. One emerging approach is antibody-peptide conjugates (APCs), which attach GLP-1-based peptides to antibodies to significantly prolong circulation time. By increasing molecular size and leveraging Fc receptor-mediated antibody recycling, APCs can substantially extend half-life while maintaining biological activity. The leading example is Amgen's MariTide (AMG 133), a GLP-1 agonist/GIPR antagonist APC that has demonstrated placebo-adjusted weight loss of 13.7% at 52 weeks and can be administered once monthly. Chinese developers are also entering the field, with Innovent advancing IBI3030, a next-generation APC that combines anti-PCSK9 antibodies with GLP-1R/GCGR/GIPR agonist activity, highlighting the potential for multi-target metabolic and cardiovascular benefits.

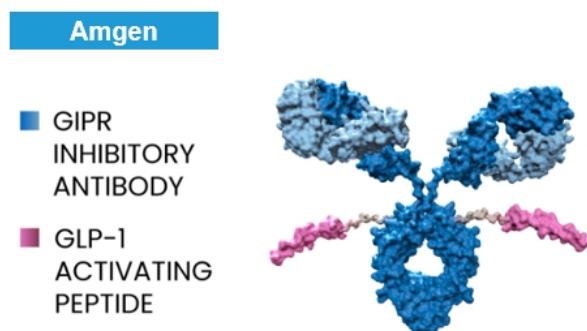
Another promising platform is circular RNA (circRNA), which may overcome some of the durability limitations of conventional mRNA-based therapies. Unlike linear mRNA, circRNAs are inherently more stable and can support prolonged protein expression, making them attractive for long-acting metabolic treatments. PegBio's CR059 uses lipid nanoparticle delivery to encode a GLP-1 analogue and is designed for monthly dosing. In preclinical studies in diabetic non-human primates, fasting plasma glucose fell by approximately 31% within two days, reached target levels within the first week, and remained controlled for about one month after a single administration. Although still in Phase 1 development, the platform illustrates how RNA therapeutics could potentially transform obesity and diabetes treatment from weekly to monthly interventions.

Gene therapy represents an even more ambitious approach. Fractyl Health is developing a beta-cell-targeted "smart GLP-1" therapy using an adeno-associated virus (AAV) vector delivered through an endoscopic ultrasound-guided procedure. Rather than repeatedly administering a GLP-1 drug, the therapy aims to convert pancreatic beta cells into a continuous source of GLP-1, with secretion linked to metabolic signals. In preclinical mouse studies, a one-time treatment generated more than 25% weight loss over 45 days, suggesting the possibility of highly durable efficacy without chronic dosing.

While these technologies remain at an earlier stage than established GLP-1 therapies, they demonstrate how the obesity field is evolving beyond simply improving weekly injections. APCs are currently the most clinically validated of the three approaches, but circRNA and gene therapy platforms offer the prospect of extending dosing intervals to monthly—or potentially achieving long-lasting therapeutic effects after a single treatment. If successful, these innovations could significantly reshape the long-term treatment paradigm for obesity and metabolic disease.

Note: Amgen is covered by Courtney Breen

EXHIBIT 17: **AMG 133, an antibody-peptide conjugate GLP-1**



Source: Company disclosure, Bernstein analysis

EXHIBIT 19: **PegBio has developed circRNA**

PegBio

LB: CLINICAL THERAPEUTICS—INCRETIN-BASED THERAPIES | JUNE 05 2026

2832-LB: CR059, a Once-Monthly circRNA-Encoded GLP-1 Analogue, Demonstrates Durable Glycemic Control in T2D Rhesus Monkeys **FREE**

YA PEI, MICHAEL MIN XU

[Check for updates](#)

Diabetes 2026;75(Supplement_1):2832-LB
<https://doi.org/10.2337/db26-2832-LB>

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Introduction and Objective: CR059 is a circular RNA-encoded GLP-1 receptor agonist delivered via lipid nanoparticles (LNP), designed to enable sustained expression and long-acting metabolic control with once-monthly subcutaneous dosing. This study evaluated the glucose-lowering efficacy of CR059 in type 2 diabetic (T2D) monkeys.

Source: ADA, Bernstein analysis

EXHIBIT 18: **Another APC example is Innovent's IBI3030**

Innovent

LB: LIPIDS | JUNE 20 2025

1886-LB: A Novel Antibody–Peptide Conjugate Targeting PCSK9, GLP-1R, GCGR, GIPR Improves Cardiovascular Risk Markers in Preclinical Study **FREE**

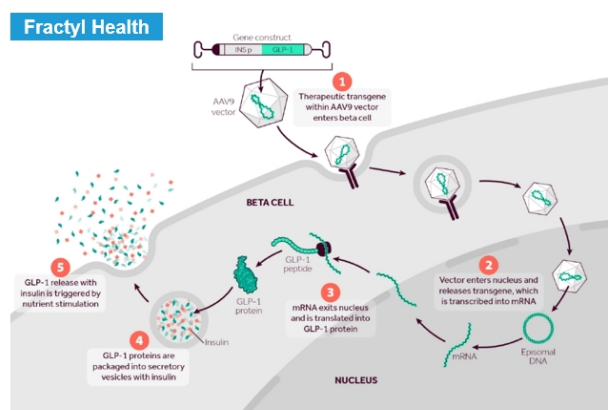
DECHENG REN, YIMING LI

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Diabetes 2025;74(Supplement_1):1886-LB
<https://doi.org/10.2337/db25-1886-LB>

Source: ADA, Bernstein analysis

EXHIBIT 20: **An ambitious therapy introduced by Fractyl Health**



Source: Fractyl Health

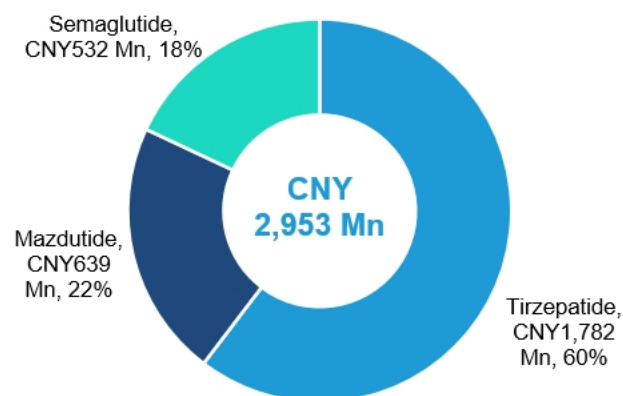
PART 5. CHINA MARKET UPDATE

GLP-1 retail demand remained robust in 1H26 despite regulatory noise around online obesity-drug sales. Online sales of the three leading GLP-1 products—tirzepatide (TZP), mazdutide (MAZ) and semaglutide (SEMA)—continued to grow strongly through 1H26, with combined online retail sales approaching CNY3 Bn based on channel tracking. In May, media reports indicated tighter controls on online sales of obesity medicines, including stricter prescription verification requirements and restrictions on internet-hospital prescriptions in certain regions. However, GLP-1 products largely remained accessible because they are also approved for type 2 diabetes. While all three products experienced modest sequential moderation in June, we think it is too early to conclude that the regulatory tightening has materially altered underlying demand trends.

Tirzepatide remained the category leader, and mazdutide was the key share gainer in 1H26. Based on our channel data, TZP generated approximately CNY1.8 Bn of retail sales during 1H26, representing about 60% share of the tracked GLP-1 market. Nevertheless, Innovent's MAZ continued to gain share against SEMA and surpassed it in monthly sales from March onward. At the current trajectory, we estimate MAZ online sales could reach approximately CNY1.3–1.5 Bn in 2026. Assuming online sales account for roughly 60–70% of total retail sales, we believe total MAZ sales remain on track to achieve the company's ~2Bn target for 2026.

EXHIBIT 21: Mazdutide recorded over CNY 600 Mn online sales in 1H26, getting 22% of share as 2nd place among Top 3 GLP1s

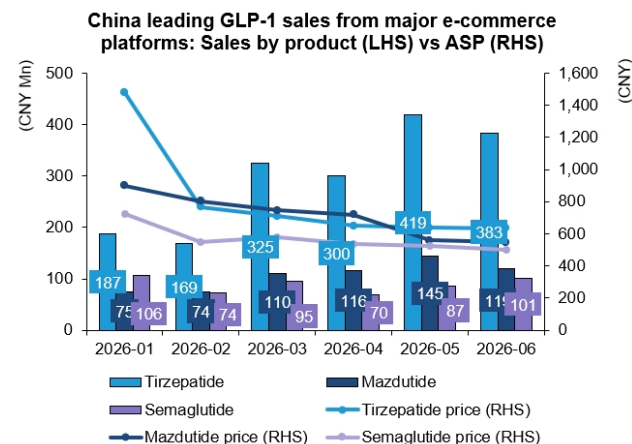
26H1 GLP-1 online sales by product



Semaglutide: include ozempic and wegovy

Source: Moojing Intelligence (Taobao + Tmall + JD), Bernstein analysis

EXHIBIT 22: Tirzepatide online sales grew the most in recent months; mazdutide had surpassed semaglutide sales since March



Semaglutide: include ozempic and wegovy

Source: Moojing Intelligence (Taobao + Tmall + JD), Bernstein analysis

We updated our China GLP-1 market model to reflect recent pricing trends and a broader treatment paradigm, with faster than-expected price erosion offset by a meaningful expansion in TAM and penetration. Based on the price reset starting from Jan. 2026, we lowered our obesity ASP (average selling price) assumption to c.CNY270/month by 2030E (from c.CNY430/month previously), with a long-term normalization toward c.CNY200/month.

In parallel, we have expanded the total addressable market (TAM) by broadening the eligible patient population. Our previous model assumed a relatively high threshold for adoption, focusing only on obese patients (BMI > 28) with at least one comorbidity. However, recent approvals support the use of weight loss therapies in a wider population, including both obese patients (with or without comorbidities) and overweight individuals (BMI btw 24–28) with at least one weight-related comorbidity. Reflecting this shift, we now include both groups in our TAM.

Alongside the expanded population, we have increased our assumptions for GLP-1 penetration. Real-world data suggest that uptake is accelerating, with U.S. penetration among overweight and obese populations already estimated at 10–20% and continuing to rise. A 2024 report further indicates that 12% of all U.S. adults have used GLP-1 therapies at some point, with approximately 6% actively using them. Based on these trends, we raise our projected penetration in the overweight/obese population from 8% to 15% by 2030E. At the same time, we refine treatment duration assumptions to better reflect observed behavior, reducing expected duration from 8 months to 3 months for weight loss and from 12 months to 10 months for type 2 diabetes. We also incorporate the growing importance of juvenile obesity as an incremental driver of demand. Obesity prevalence is projected to rise more rapidly among children than adults, increasing from around 10% today to 18% by 2035E, compared with adult prevalence rising from 14% to 18% over the same period.

Overall, while faster price erosion presents a headwind, it is more than offset by the expansion in TAM and higher expected GLP-1 penetration. We therefore continue to project that China GLP-1 market will reach c.CNY85 Bn by 2030E. Among all key assumptions, penetration in the overweight/obese population—now forecast at 15%—remains the most critical and uncertain variable, particularly given the low starting base of around 1% today.

EXHIBIT 23: **Online channel price tracker for obesity**

I. Obesity - Online channel						
Monthly price tracker (price per package * # pack per month)						
Product	mg/package	# pack per month	1/21/2026	3/31/2026	6/8/2026	7/22/2026
Semaglutide						
Ozempic						
0.25mg/wk	2.0	0.5	185	168	163	160
0.5mg/wk	2.0	1.0	369	335	326	320
1.0mg/wk	4.0	1.0	550	544	519	539
Wegovy						
0.25mg/wk	1.0	1.0	211	188	171	231
0.5mg/wk	2.0	1.0	358	300	266	325
1.0mg/wk	4.0	1.0	523	461	420	552
1.7mg/wk	6.8	1.0	908	750	621	989
2.4mg/wk	9.6	1.0	1204	871	797	1286
Tirzepatide						
2.5mg/wk	10	1.0	530	449	404	459
5.0mg/wk	20	1.0	769	714	669	754
7.5mg/wk	30	1.0	1089	959	914	1049
10.0mg/wk	40	1.0	1339	1219	1154	1299
Mazdutide						
2.0mg/wk	8	1.0	760	605	490	648
4.0mg/wk	16	1.0	1362	1181	910	1248
6.0mg/wk	24	1.0	1866	1643	1280	1789

Source: JD.com, Bernstein analysis

EXHIBIT 24: Semaglutide and tirzepatide are included in NRDL in China, and we think mazdutide will be included starting from 2027E

II. T2D - NRDL				
Monthly price tracker (price per package * # pack per month)				
Product	mg/package	# pack per month	2024 NRDL price	2026 (NRDL) price
Semaglutide				
Ozempic				
0.25mg/wk	2.0	0.5	211	194
0.5mg/wk	2.0	1.0	421	388
1.0mg/wk	4.0	1.0	716	660
Tirzepatide				
2.5mg/wk	10	1.0	\	324
5.0mg/wk	20	1.0	\	551
7.5mg/wk	30	1.0	\	752
10.0mg/wk	40	1.0	\	937
Mazdutide				
2.0mg/wk	8	1.0	\	490
4.0mg/wk	16	1.0	\	910
6.0mg/wk	24	1.0	\	1280

Mazdutide price in the chart is from online channels, given the product has not been included in NRDL

Source: NRDL list, Bernstein analysis

2027 marks the beginning of a major GLP-1 launch wave. Following the initial launches of semaglutide (Ozempic, Wegovy, and Rybelsus), tirzepatide (Mounjaro), and mazdutide (Xinermei), China GLP-1 market is expected for a concentrated wave of new entrants in 2027E, including Hengrui's HRS-9531, Hansoh's HS-20094, orforglipron (oral GLP-1) and the first wave of semaglutide biosimilars following patent expiry. The pipeline is expected to broaden further in 2028E and beyond with the potential launches of Cagrisema, oral semaglutide (Wegovy pill), oral HRS-9531, GZR18, BGM0504, and other late-stage candidates, expanding both injectable and oral treatment options. We believe this launch wave will likely accelerate market penetration and patient access while driving the transition to a competitive multi-player landscape. As competitive intensity grows, differentiation is likely to shift beyond glycemic and weight-loss efficacy toward factors such as oral convenience, body composition benefits, and pricing.

EXHIBIT 25: Following the launch of Mazdutide, domestic GLP-1 market is expected to see a concentrated wave of new product launch starting in 2027 and beyond

	2024	2025	2026E	2027E	2028E or later
Injections	Lilly Tirzepatide (Mounjaro) ▲ Approved in China (May'24 T2D; Jul'24 obesity) ▲ Official commercial launch in China in Jan. 2025 Novo Nordisk Semaglutide (Ozempic & Wegovy) ▲ Approved in China (Apr'21 T2D; Jun'24 obesity) ▲ Launched both Ozempic and Wegovy in China	Innovent Mazdutide (IBI362) ▲ Approved in China (Jun'25 obesity) (Nov'25 9mg filed) ▲ Approved in China (Sept'25 T2D) PegBio Visepegenatide ▲ Approved in China (Nov'25 T2D; obesity in Ph1/2)	SCIWIND Ecnoglutide ▲ Approved in China (Jan'26 T2D) (Mar'26 obesity)	Hengrui Ribupatide (HRS-9531) ▲ Filed NDA in Sep'2025 in China, est. launch in 2027E Hansoh HS-20094 ▲ Filed NDA in Jun'2026, est. launch in 2027E Semaolotide biosimilar ▲ China patent expiry in 2026; expected to approve in 2027E	Novo Nordisk CagriSema ▲ Est. to be approved in China in 2028E+ Gan & Lee GZR18 ▲ Now in Ph3, est. launch in 2028E+ BrightGene BGM0504 ▲ Now in Ph3, est. launch in 2028E+
Oral	Novo Nordisk Semaglutide (Rybelsus) ▲ Approved in China (Jan'24 T2D)			Lilly Orforglipron ▲ NDA submitted in China for both T2D and obesity, est. launch in 2027E	Novo Nordisk Semaglutide (Wegovy pill) ▲ Phase 3 OASIS 3 completed in China, expected approval in 2028E+ Hengrui Oral ribupatide ▲ China Phase 2 trial ongoing, est. launch in 2028E+ Hengrui HRS-7535 ▲ Now in Ph3, est. launch in 2028E+ Incentage VCT220 ▲ Now in Ph3, est. launch in 2028E+

PegBio, Incentage, and Gan & Lee are not covered

Source: Company disclosures, ClinicalTrials.gov, Pharmcube, Bernstein estimates and analysis

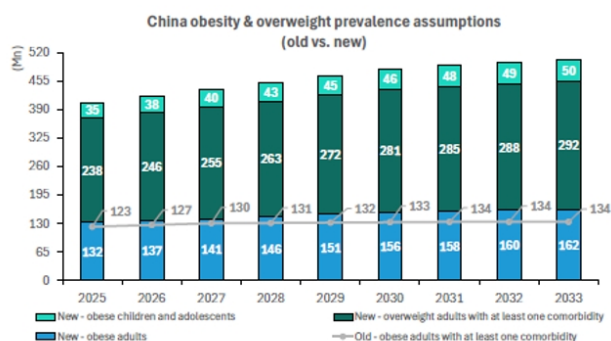
EXHIBIT 26: Key assumptions updates for 2030E China GLP-1 market

Obesity and overweight market segment		
Metric	2030E (old)	2030E (new)
Overweight and obese prevalence (Mn)	133*	483
GLP-1 treated patients (Mn)	18	79
Duration of treatment (m)	8 (obesity); 2 (non-obesity)	3 (obesity); 1 (non-obesity)
GLP-1 patient months (Mn)	101	224
Monthly ASP of GLP-1 (CNY/m)	433	268
China GLP-1 market size - obesity (CNY Bn)	44	60
Type 2 diabetes (T2D) market segment		
T2D prevalence (Mn)	177	127
GLP-1 treated patients (Mn)	15	9
Duration of treatment (m)	12	10
GLP-1 patient months (Mn)	185	95
Monthly ASP of GLP-1 (CNY/m)	223	265
China GLP-1 market size - T2D (CNY Bn)	41	25
Total market size		
China GLP-1 market size - total (CNY Bn)	85	85

a. We did not include overweight in previous model

Source: Company disclosures, DIABETES OBESITY AND METABOLISM, NIH, MedNexus, Bernstein analysis and estimates

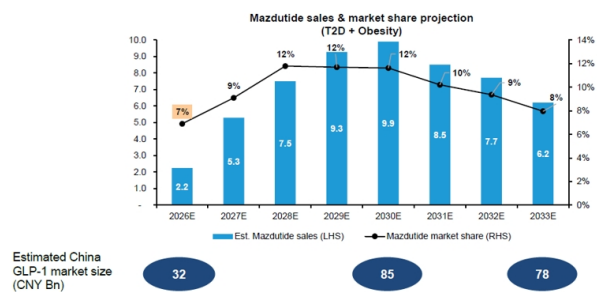
EXHIBIT 27: Prevalence assumptions of China obesity & overweight old vs. new



a. We did not include overweight in previous model

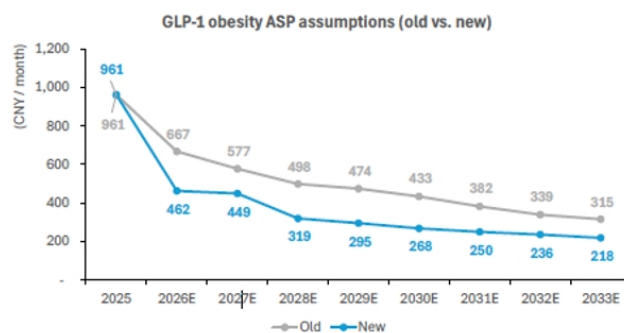
Source: Company disclosures, DIABETES OBESITY AND METABOLISM, NIH, MedNexus, Bernstein analysis and estimates

EXHIBIT 28: Updated sales trajectory for mazdutide



Source: Bernstein estimates and analysis

EXHIBIT 29: China obesity market ASP assumptions for GLP-1s old vs new



Source: Company disclosures, NHSA, Pharmcube, JD, Bernstein analysis and estimates

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